

Change of Basis and Linear Transformations

Generalizing this discussion, assume that

$$B = \{v_1, v_2, \dots, v_n\} \quad \text{and} \quad B' = \{u_1, u_2, \dots, u_n\}$$

are two ordered bases for R^n . If x is a vector in R^n and

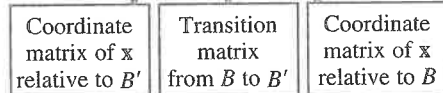
$$[x]_B = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \quad \text{and} \quad [x]_{B'} = \begin{bmatrix} d_1 \\ d_2 \\ \vdots \\ d_n \end{bmatrix}$$

are the coordinate matrices of x relative to B and B' , then the **transition matrix P from B' to B** is the matrix P such that

$$[x]_B = P[x]_{B'}.$$

The next theorem tells you that the transition matrix P is invertible and its inverse is the **transition matrix from B to B'** . That is,

$$[x]_{B'} = P^{-1}[x]_B.$$



THEOREM 4.20 The Inverse of a Transition Matrix

If P is the transition matrix from a basis B' to a basis B in R^n , then P is invertible and the transition matrix from B to B' is P^{-1} .

THEOREM 4.21 Transition Matrix from B to B'

Let

$$B = \{v_1, v_2, \dots, v_n\} \quad \text{and} \quad B' = \{u_1, u_2, \dots, u_n\}$$

be two bases for R^n . Then the transition matrix P^{-1} from B to B' can be found by using Gauss-Jordan elimination on the $n \times 2n$ matrix $[B' \ B]$, as shown below.

$$[B' \ B] \rightarrow [I_n \ P^{-1}]$$

Transformation Matrix for Nonstandard Bases

Let V and W be finite-dimensional vector spaces with bases B and B' , respectively, where

$$B = \{v_1, v_2, \dots, v_n\}.$$

If $T: V \rightarrow W$ is a linear transformation such that

$$[T(v_1)]_{B'} = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}, [T(v_2)]_{B'} = \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}, \dots, [T(v_n)]_{B'} = \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

then the $m \times n$ matrix whose n columns correspond to $[T(v_i)]_{B'}$

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

is such that $[T(v)]_{B'} = A[v]_B$ for every v in V .

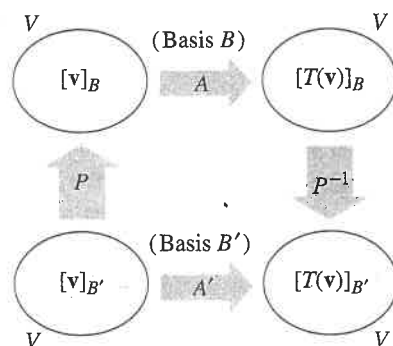


Figure 6.4

1. Matrix for T relative to B : A
2. Matrix for T relative to B' : A'
3. Transition matrix from B' to B : P
4. Transition matrix from B to B' : P^{-1}

Note that in Figure 6.4, there are two ways to get from the coordinate matrix $[v]_{B'}$ to the coordinate matrix $[T(v)]_{B'}$. One way is direct, using the matrix A' to obtain

$$A'[v]_{B'} = [T(v)]_{B'}.$$

The other way is indirect, using the matrices P , A , and P^{-1} to obtain

$$P^{-1}AP[v]_{B'} = [T(v)]_{B'}.$$

This implies that $A' = P^{-1}AP$.